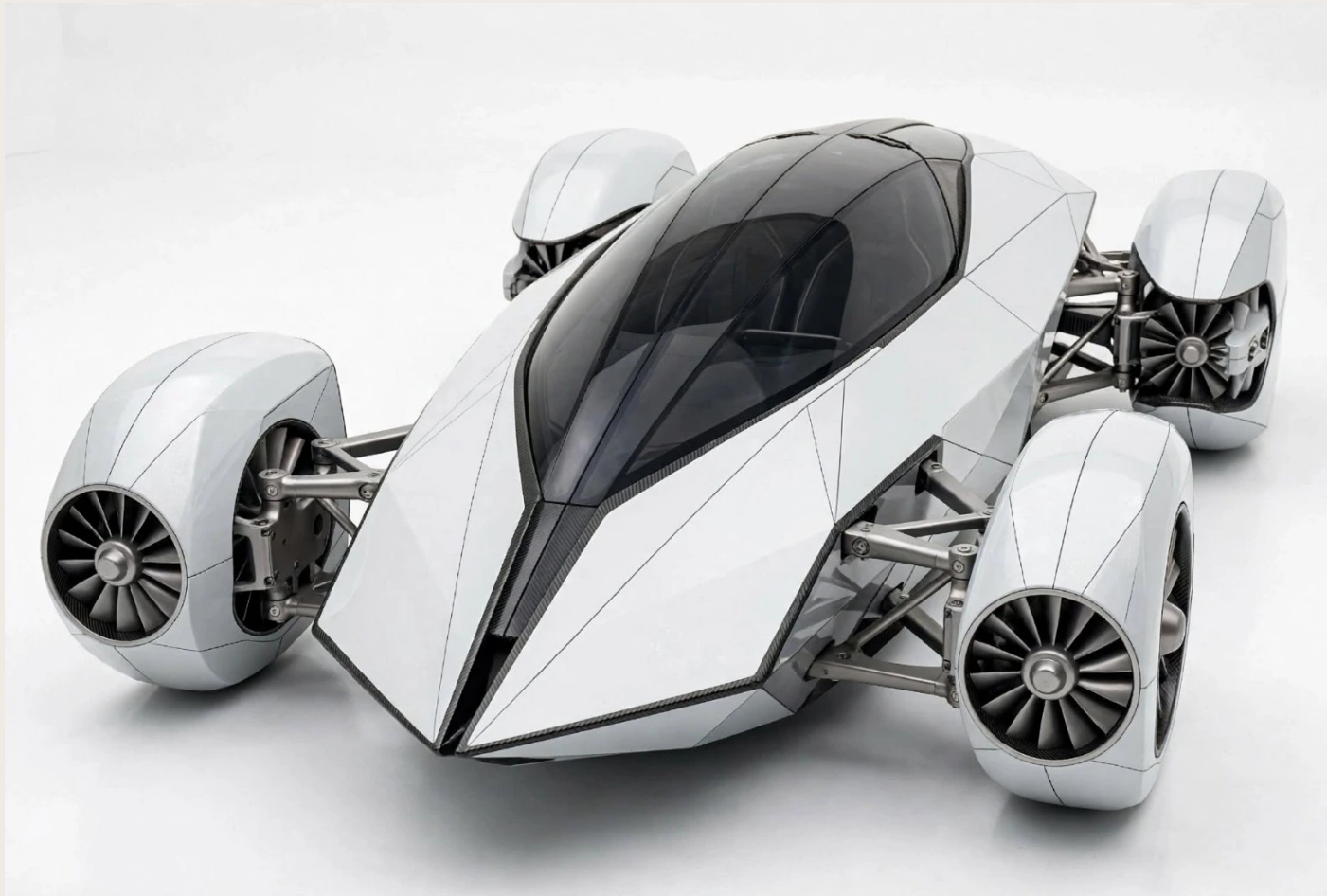
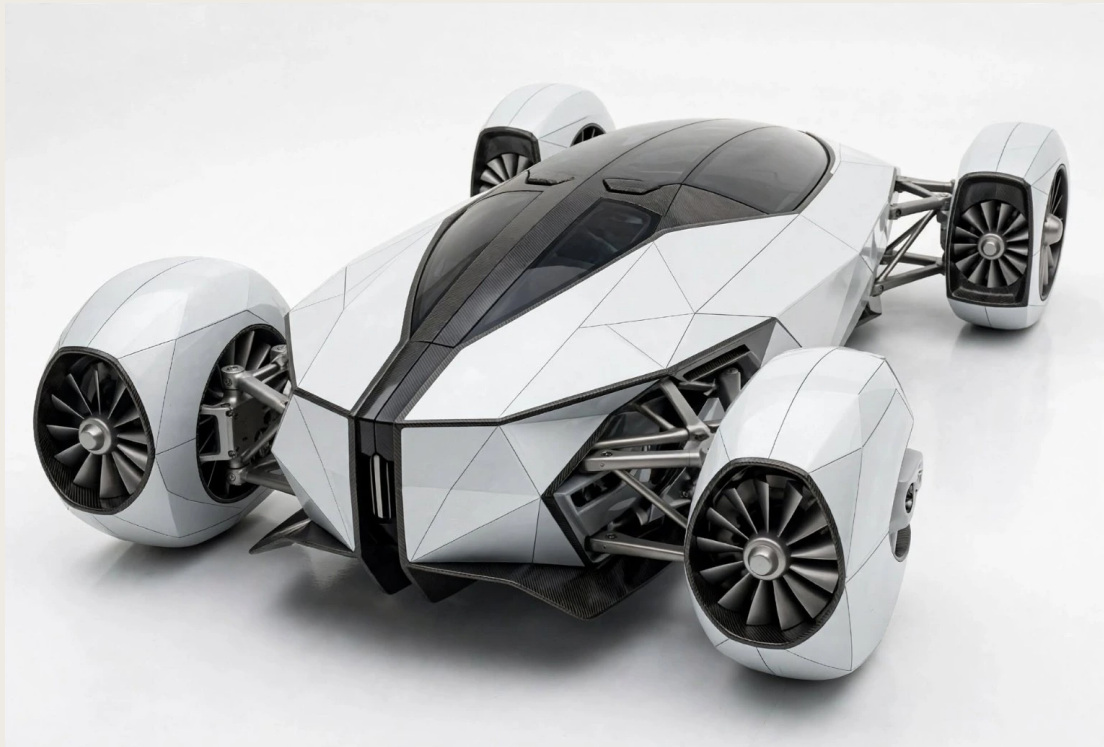


Poised between ground and sky.



AEROVANE V4 — a two-seat capsule and four-pod propulsion study. Supplied reference rendering.

Four stations. One fluid form.



A pointed central capsule, continuous black canopy and exposed triangulated links establish a lightweight, mechanical character. Four independent fairings give the platform its wide, planted stance.

AEROVANE combines the ambition of air mobility with the directionality of a vane. Its four-pod logo surrounds a V-shaped central keel. Pearl shells, dark structure and satin rotors make each system visually legible.

Reference images establish visual intent. They do not provide measured geometry, material grades or operating specifications.

Keep the casing still. Let the rotor move.

01 MAIN FRAMEWORK

Central shell, glossy black opaque canopy and mechanical links.
Eight independent material zones cover all visible surfaces.

02–05 FOUR POD HOUSINGS

Stationary front-left, front-right, rear-left and rear-right fairings
partitioned spatially from the original merged mesh.

06 ROTATABLE FAN ASSEMBLY

Four rebuilt rotor disks with separate X-axis pivots, 14 swept blades
per rotor and counter-rotation. Fan surfaces are reconstructed for
clean motion; they are not aerodynamically validated.

PRESENTATION SYSTEM

Classic non-PBR material controls; ivory, blueprint and graphite
sketches; perspective ambient occlusion; studio and nocturne
environments. Pure glossy black windshield by default.

MOTION & SOUND

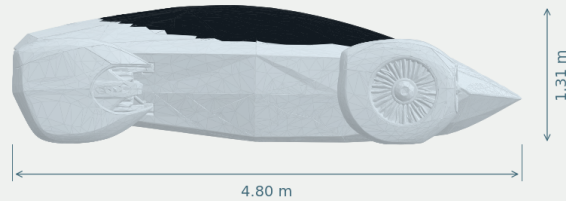
Gradual spool, spring-damped hover settling, yaw, micro vibration
and optional device haptics. Synthesized fan airflow, motor tones,
beeps and hydraulic cues require user sound activation.

Displayed fan speed and altitude are presentation controls, not a validated flight
simulation.

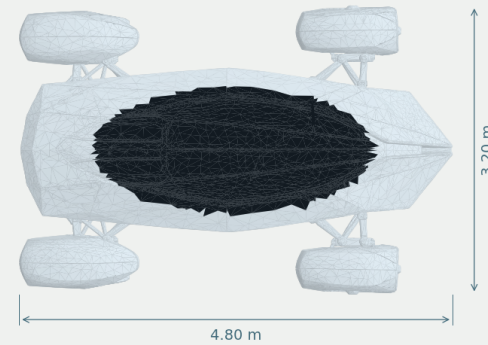
Dimensioned from an explicit assumption.

AEROVANE V4 / CONCEPT CADD PROJECTIONS

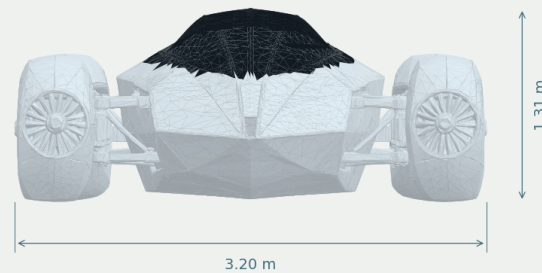
LEFT ELEVATION



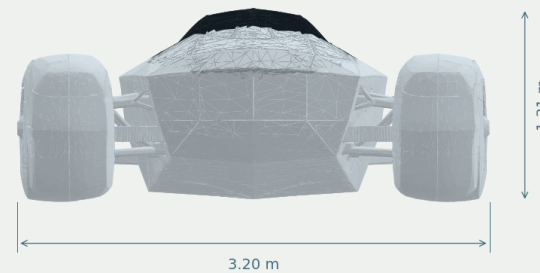
PLAN



FRONT ELEVATION



REAR ELEVATION



ASSUMED LENGTH 4.80 m • PROPORTIONAL DIMENSIONS ONLY • NOT FOR FABRICATION

4.80 m length assumed. 3.20 m width and 1.31 m height derived from mesh proportions. Fan diameter approximately 0.66 m.

A human centre. An exposed mechanism.



VISIBLE IN THE REFERENCE

Two seats, split instrument consoles, a lifting canopy and triangulated pod links. These are observations from the supplied imagery.

PROPOSED DEVELOPMENT

Composite exterior fairings over a load-bearing chassis; metallic hardpoints and pod attachments; closed-loop electric drive. Canopy hinges, emergency egress, rotor containment and landing loads require engineering.

Source geometry shows lateral fan axes. This alone cannot establish vertical lift. Tilting pods or a downward-flow architecture would need to be developed.

Compact disks. A demanding power budget.

ILLUSTRATIVE SIZING CASE

850 kg assumed gross mass. Four approximately 0.66 m fans.
Candidate motor budget: 4 x 200 kW peak. Candidate battery: 60–80 kWh. All are design assumptions, not measured or proven specifications.

FIRST-ORDER HOVER CHECK

$A = 4 \pi (D/2)^2 = 1.36 \text{ m}^2$. $T = m g = 8338 \text{ N}$. Ideal power $P = T^{3/2} / \sqrt{2 \rho A} = 418 \text{ kW}$ at $\rho = 1.225 \text{ kg/m}^3$.

Assuming 0.65 figure of merit and 0.92 drive efficiency gives 699 kW before installation losses. 60 kWh at 80% usable energy gives about 4.1 minutes before reserves and auxiliary loads.

KEY LIMITATION

This calculation assumes vertical thrust from all disks. The lateral-axis geometry shown does not satisfy that assumption. The result is a sizing challenge to resolve, not evidence of flight capability.

AVAILABLE TECHNOLOGY / OPEN WORK

Electric motors and inverters, battery systems, inertial sensing and closed-loop flight control are available technology categories. Their integration, cooling, fault tolerance, rotor containment and vehicle stability remain unvalidated.

NASA tiltduct research explores thrust-vectoring and disk-loading tradeoffs. A low disk-loading alternative or larger rotors should be compared before freezing this form.

An inspectable design study.

DELIVERED

Interactive website; embedded six-group model; four animated rotor pivots; surface configurator; reference gallery; dimensioned orthographic plate; vector projection SVG; this design dossier.

GEOMETRY STATUS

The source was one merged mesh. Spatial partitioning preserves the main silhouette, with possible boundary seams. Outboard rotor disks were rebuilt and given local axes. No original hinge or drive mechanism was supplied.

TECHNOLOGY REFERENCES

NASA: Conceptual Design of a Tiltduct Reference Vehicle for Urban Air Mobility. <https://ntrs.nasa.gov/citations/20210025911>

NASA: Urban Air Mobility research and disk-loading considerations. <https://rotorcraft.arc.nasa.gov/Research/Programs/Urban%20Air%20Mobility.html>

VALIDATION

Model structure, animation axes, JavaScript syntax and asset paths checked. Browser visual QA unavailable. No airworthiness, endurance, noise, structural or manufacturing certification is claimed.